



Practice all the questions:

1. Find the Taylor's series expansion of the function $f(z) = \frac{1}{z^2 - z - 6}$ about $z = 1$.

Answer: $f(z) = \frac{-1}{6} + \frac{1}{5} \left(\frac{1}{2^2} - \frac{1}{3^2} \right) (z-1) - \left(\frac{1}{2^3} + \frac{1}{3^3} \right) (z-1)^2 + \left(\frac{1}{2^4} - \frac{1}{3^4} \right) (z-1)^3 + \dots$

2. Find the Taylor's series expansion of the function $f(z) = \frac{z+1}{(z-3)(z-4)}$ about $z = 2$.

Answer: $f(z) = 4(1 + (z-2) + (z-2)^2 + \dots) - \frac{5}{2} \left(1 + \frac{z-2}{2} + \frac{(z-2)^2}{4} + \dots \right)$

3. Find the Laurent series expansion of the function $f(z) = \frac{z}{z^2 - 4z + 3}$ in the region $1 < |z| < 3$.

Answer: $f(z) = \frac{-1}{6} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \dots \right) - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right)$

4. Find the Laurent series expansion of the function $f(z) = \frac{7z-2}{(z+1)z(z-2)}$ in the region $1 < |z+1| < 3$.

Answer: $f(z) = \frac{-3}{z+1} + \frac{1}{z+1} \left(1 + \frac{1}{z+1} + \frac{1}{(z+1)^2} + \dots \right) - \frac{2}{3} \left(1 + \frac{z+1}{3} + \frac{(z+1)^2}{9} + \dots \right)$